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RESEARCH ARTICLE



Evaluation of the translational and rotational diffusion coefficients of a cubic particle (for the application to Brownian dynamics simulations)

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ABSTRACT

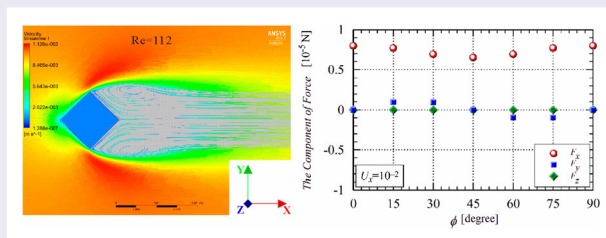
By estimating the force and torque acting on the cube for the two cases of a uniform flow field and a rotational flow field, we have discussed whether or not there is a coupling between the translational and the rotational motion. From the characteristics of the friction coefficients, we may understand that there is no coupling between the translation motion and the rotational motion in the situation of the Reynolds number being sufficiently smaller than unity. In contrast, in the case of a non-slow flow field with the Reynolds number larger than unity, the coupling characteristics of the motion of a cube is certainly recognised and therefore the interaction with the ambient fluid is characterised by a variety of friction coefficients including friction coefficients that relate the forces acting on the cube to the angular velocities of the rotational motion. Hence, the employment of these translational and rotational diffusion coefficients for a cube enables the implementation of Brownian dynamics simulations for a suspension composed of cubic particles in order to analyse the dynamic characteristics of a cubic particle suspension.

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KEYWORDS

Diffusion coefficient; friction coefficient; Stokes flow; resistance matrix; Brownian dynamics simulation



Highlights

- (1) We have considered a flow problem around a cube in order to numerically clarify the characteristics of the translational and rotational friction or diffusion coefficients.
- (2) In a slow flow field the motion of the cube need only to be characterised by two friction coefficients, i.e. the translational and rotational friction coefficients.
- (3) In the case of a non-slow flow field, the coupling characteristics between the translational motion and the rotational motion are recognised.
- (4) Employment of these diffusion coefficients enables the implementation of Brownian dynamics simulations for a suspension composed of cubic particles.

1. Introduction

A magnetic particle suspension has a hopeful potential for applications in a variety of fields such as medical engineering and environmental resource engineering in addition to the more established applications in fluid engineering and magnetic recording material engineering. The main target for application in medical

engineering is a drug delivery system where magnetic composite particles loading anti-cancer medicines are magnetically guided to a cancer or tumour site [1] and a magnetic hyperthermia therapy where heat production is generated by magnetic particles from the relaxation mechanism of the magnetic moments in an alternating magnetic field [2,3]. In environmental resource

engineering, multi-functionalized magnetic particles are used for adsorbing noble metals from sea water or hazard metal ions from industrial waste water and then the particles are captured via their motion in a gradient magnetic field [4,5].

In the field of fluid engineering, the main targets of application are mechanical dampers and actuators that make use of the magnetorheological effect [6,7]. The magnetorheological effect here generally implies an increase in an apparent viscosity due to particle alignment or the formation of particle clusters in the situation of an external magnetic field, leading to a large resistance to the flow of a particle suspension. Hence the magnetorheological effect in a suspension is strongly dependent on the geometrical shape and the magnetic characteristics of the magnetic particles [8–10]. For example, ferromagnetic rod-like particles magnetised in the major axis direction are expected to exhibit a significantly larger magnetorheological effect in comparison to spherical magnetic particles. In addition, the magnetorheological effect in rod-like particles is dependent in a complex manner on the internal structure of particle aggregates, where a phase change in the internal structure between raft-like and chain-like clusters arises due to the influence of an external magnetic field [11]. In contrast, rod-like hematite particles that are magnetised in the minor axis direction exhibit magnetorheological characteristics completely different from those of the above-mentioned ferromagnetic rod-like particles. It has been observed that they exhibit a negative magnetorheological effect with an apparent viscosity decrease in certain circumstances of an applied magnetic field, which had been predicted from theoretical analyses and Brownian dynamic simulations [12] and verified by experimental study [13]. A suspension composed of cubic magnetic particles is expected to exhibit a more complex magnetorheological characteristic effect that may arise from the formation of a more rigid aggregate structure based on the geometry of a face-to-face contact between the neighbouring constituent particles in a cluster [14]. As will be described below, a magnetic cubic particle suspension exhibits a complex internal structure, phase change and orientational characteristics that are dependent on the strength and direction of an applied magnetic field.

In order to elucidate the magnetorheological characteristics on a microscopic scale, there are several particle-based simulation techniques that may be employed such as molecular dynamics, Brownian dynamics, lattice Boltzmann, dissipative particle dynamics and multi-particle collision dynamics [15,16]. Among these simulation techniques, the Brownian dynamics method may be the more straightforward for a particle suspension, where the particle Brownian motion may be accurately activated

without introduction of the additional technique that is necessary in the lattice Boltzmann method, although the multi-body hydrodynamic interactions among particles are not taken into account in Brownian dynamics. For this reason, we here address the Brownian dynamics method as a simulation tool for a magnetic cubic particle suspension. In Brownian dynamics simulations it is required to treat the translational and rotational Brownian motion of magnetic particles [16]. In contrast to a suspension of spherical particles, particles with a non-spherical shape such as rod-like or cube-like require a more complex treatment in respect to the equations of motion as they require the inclusion of particle translational and rotational friction coefficients or diffusion coefficients. In the case of axisymmetric rod-like particles, the translational motion is decomposed into the motion in the major and the minor axis directions, and the rotational motion is treated in a similar manner. The translational and rotational diffusion coefficients that are required in simulations may be derived by modelling an axisymmetric particle such as a circular or a spheroid whose diffusion coefficients are given by well known mathematical expressions [16]. In the case of cubic particles which are not axisymmetric we are required to develop a simulation technique for a cubic particle suspension with appropriate diffusion coefficients. A success in the development of a Brownian simulation technique for a cubic particle suspension would be a useful contribution to the expansion of studies on their dynamic properties which may lead to the development of new applications for mechanical dampers and actuators.

One of application of magnetic cubic particles is in the field of surface modification technology and many researchers have been actively investigating the phenomenon of the deposition of magnetic cubic particles on a material plane surface [17–20]. In this line of research, the characteristics of the internal structure of aggregates, including a phase change and orientational features found in a cubic magnetic particle suspension have been investigated by means of 2D Monte Carlo simulations [21]. From this study, where an external magnetic field is used for controlling the above-mentioned physical characteristics, we understand that an applied magnetic field by itself cannot highly control the characteristics of the attachment and aggregate structure on a material surface to a sufficient degree. One possible approach for making this technology more accurate may be the application of a new concept whereby in addition to the control exercised by magnetic field, the magnetic particles are electrically charged and an applied electric field is also employed for controlling the attachment and aggregate of structures. From these considerations, it is evident that the development of a particle-based simulation method which is

able to treat particle Brownian motion is indispensable for analysing the dynamic characteristics of a suspension composed of magnetic cubic particles.

From this background, in the present study, we consider the translational and rotational diffusion coefficients of a cubic particle that are indispensable for evaluating the dynamic properties of a suspension when conducting Brownian dynamics simulations. In concrete, we have analysed the flow field around a cube via commercial software ANSYS CFX in order to evaluate the force and torque acting on the cube, which gives rise to the translational and rotational diffusion coefficients. These results are used for discussing how the translational and the rotational motion have a coupling relationship and then we show the equations for the Brownian dynamics simulations for a suspension composed of magnetic cubic particles.

2. Evaluation procedure of the friction coefficients by a numerical analysis method

We first consider the motion of a colloidal particle with a general geometrical shape in a quiescent flow field. The force $\mathbf{F}$ and the torque $\mathbf{T}$ acting on the ambient fluid by the particle are related to the fluid velocity $\mathbf{v}$ and the angular velocity $\boldsymbol{\omega}$ as

$$\begin{pmatrix} \mathbf{F} \\ \mathbf{T} \end{pmatrix} = \mathbf{R} \cdot \begin{pmatrix} \mathbf{v} \\ \boldsymbol{\omega} \end{pmatrix}, \quad (1)$$

in which $\mathbf{R}$ is the resistance matrix, expressed as

$$\mathbf{R} = \begin{pmatrix} \mathbf{R}_a & \mathbf{R}_c \\ \mathbf{R}_b & \mathbf{R}_d \end{pmatrix}, \quad (2)$$

Equation (1) is rewritten using Eq. (2) as

$$\begin{cases} \mathbf{F} = \mathbf{R}_a \cdot \mathbf{v} + \mathbf{R}_c \cdot \boldsymbol{\omega} \\ \mathbf{T} = \mathbf{R}_b \cdot \mathbf{v} + \mathbf{R}_d \cdot \boldsymbol{\omega} \end{cases}. \quad (3)$$

We now describe the physical meaning of each component matrix of the resistance matrix $\mathbf{R}$. The matrix $\mathbf{R}_a$ relates the force acting on the ambient fluid to the translational velocity and the matrix $\mathbf{R}_d$ relates the torque to the angular velocity. The matrices $\mathbf{R}_b$ and $\mathbf{R}_c$ imply the rate of the coupling between the translational motion and the rotational motion, where the situation of $\mathbf{R}_c = \mathbf{0}$ implies that the rotational motion does not induce a translational motion and similarly the situation of $\mathbf{R}_b = \mathbf{0}$ implies that the rotational motion is not influenced by the translational motion. Therefore, if the relationship of $\mathbf{R}_b = \mathbf{R}_c = \mathbf{0}$ is satisfied, then there is no coupling between the translational motion and the rotational motion of the cube. Hence, if the resistance component matrices are known in the form of mathematical expressions, the force and the torque acting on

the cube may be evaluated, which ensures the viability of Brownian dynamics simulations for a suspension composed of cubic particles. In the case of cubes, it is known that the matrices $\mathbf{R}_a$ and $\mathbf{R}_d$ in the situation of low Reynolds numbers much smaller than unity are diagonal with a common component ξ_{cube}^T and ξ_{cube}^R which are the translational and the rotational friction coefficient of a cube, respectively [22]. However, the coupling characteristics of the translational and the rotational motion of a cube and the correct values of the friction coefficients ξ_{cube}^T and ξ_{cube}^R were not addressed in Brenner's paper [22].

In the present study we analyse a flow field around a fixed cube using the commercial software ANSYS CFX to clarify the characteristics of the resistance component matrices in a variety of situations in respect to a particle configuration relative to a uniform flow past the cube. The governing equation to be solved is the Navier-Stokes equation with the continuity equation. In the following, we address a flow past a sphere in order to understand the derivation procedure of the friction coefficients or diffusion coefficients from the flow field analysis. A similar derivation procedure is employed for evaluating the diffusion coefficients for the present cubic particle. If we focus on a slow flow problem past a fixed sphere with the Reynolds number being much smaller than unity i.e. $Re \ll 1$, the relationship between the force $\mathbf{F}$ acting on the ambient fluid and the particle translational velocity $\mathbf{v}$ and between the torque $\mathbf{T}$ and the angular velocity $\boldsymbol{\omega}$ is written from Stokes's resistance formula as

$$\mathbf{F} = \xi_{sphere}^T \mathbf{v}, \quad (4)$$

$$\mathbf{T} = \xi_{sphere}^R \boldsymbol{\omega}, \quad (5)$$

where ξ_{sphere}^T and ξ_{sphere}^R are the translational and the rotational friction coefficient which are expressed as $\xi_{sphere}^T = 6\pi\eta a$ and $\xi_{sphere}^R = 8\pi\eta a^3$, respectively, with liquid viscosity η for a sphere with radius a . Hence, the translational and the rotational diffusion coefficients, D_{sphere}^T and D_{sphere}^R , are obtained from the relationships:

$$D_{sphere}^T = \frac{kT}{\xi_{sphere}^T}, \quad D_{sphere}^R = \frac{kT}{\xi_{sphere}^R}, \quad (6)$$

in which k is Boltzmann's constant and T is the liquid temperature. Since there is no coupling between the translational and the rotational motion for the case of a sphere, the resistance component matrices $\mathbf{R}_b$ and $\mathbf{R}_c$ in Eq. (3) give rise to the relationship of $\mathbf{R}_b = \mathbf{R}_c = \mathbf{0}$, whereas the resistance component matrices $\mathbf{R}_a$ and $\mathbf{R}_d$ are the diagonal matrices with a common component ξ_{sphere}^T and ξ_{sphere}^R , respectively.

If the friction coefficients or the diffusion coefficients are not known in the form of analytical expressions or numerical data, they may be evaluated by a numerical analysis method in a following manner. For simplicity, we focus on how the translational friction coefficients are evaluated from numerical flow simulations. A sphere is fixed at a certain position in a uniform flow field $\mathbf{U}$ and the flow field is solved by a numerical analysis. The numerical solution of the flow enables the evaluation of the force $\mathbf{F}^P = (F_x^P, F_y^P, F_z^P)$ acting on the sphere. If the uniform flow field is applied in the x -direction as $\mathbf{U} = (U_x, 0, 0)$, the translational friction coefficient $\xi_{sphere}^T = 6\pi\eta a$ is obtained from Eq. (3) using F_x^P and U_x as

$$\xi_{sphere}^T = \frac{F_x^P}{U_x}. \quad (7)$$

In a similar procedure, the rotational flow field $\boldsymbol{\Omega} = (\Omega_x, 0, 0)$ about the x -axis yields the rotational friction coefficient ξ_{sphere}^R as

$$\xi_{sphere}^R = \frac{T_x^P}{\Omega_x}. \quad (8)$$

The diffusion coefficients D_{sphere}^T and D_{sphere}^R will then finally be obtained from the relationship in Eq. (6). As will be recognised, the implementation of Brownian simulations requires the values of the translational and rotational friction coefficients or diffusion coefficients.

In the present case of a cubic particle, the relationship of $\mathbf{R}_b = \mathbf{R}_c = \mathbf{0}$ is not necessarily satisfied but rather the resistance component matrices are expected to be dependent on the orientational configuration relative to a uniform flow and also on the Reynolds number.

Finally, it is noted that the Reynolds number Re is defined as

$$Re = \frac{U_0 d}{\nu}, \quad (9)$$

in which U_0 is the representative velocity, d is the side length of the cube and ν is the kinematic viscosity.

3. Assignment of the simulation parameters

The following values of the system parameters were employed in performing the numerical simulations both for a uniform flow past a cube and for a rotational flow about the x -axis direction. We treat a flow problem for water with the physical properties of viscosity

$\eta = 889.9 \times 10^{-6}$ Pa·s, density $\rho = 997$ kg/m³ and kinematic viscosity $\nu = \eta/\rho = 8.926 \times 10^{-7}$ m²/s relating to a temperature of 25°C. The side length of a cube is taken as $d = 0.01$ m and is fixed at the origin of the xyz -coordinate system. It is noted that the unit mesh size is required to be taken at a significantly small value in comparison to the side length of a cube, therefore, the size of a cube can be taken as any value provided the lattice system can be taken sufficiently fine. For the case of a uniform flow in the x -axis direction, the simulation box is rectangular and for the case of a rotational flow about the x -axis direction it is cylindrical. In both the cases the unit mesh size is taken as 1×10^{-4} m. As usual a no-slip boundary condition is employed at the surface of the cube, and a no-gradient condition is adopted at the outer boundary surface. A sufficiently large sized simulation region, $(L_x, L_y, L_z) = (1 \text{ m}, 0.5 \text{ m}, 0.5 \text{ m})$, is employed in order to ensure the independence of the simulation results, where it has been confirmed that a simulation region with double the size yields at most only a 1% deviation in the results. The inflowing uniform velocity is set as $U_x = 1 \times 10^{-2} \sim 1 \times 10^{-6}$ m/s and the angular velocity is taken as $\Omega\Omega_x = 1 \sim 1 \times 10^{-6}$ rad/s, which give rise to situations with a variety of different Reynolds numbers. That is, the uniform flow with $U_x = 1 \times 10^{-6}$ m/s gives rise to $Re = 1.12 \times 10^{-2}$ that is evaluated from Eq. (9) with $U_0 = U_x$, and similarly the rotational flow with $\Omega\Omega_x = 1 \times 10^{-6}$ yields $Re = 1.12 \times 10^{-4}$ with $U_0 = \Omega\Omega_x d$. Both these cases satisfy the condition of $Re \ll 1$ for the situation of a slow flow. Table 1 shows the validity of the present numerical simulations where the numerical results are compared with the corresponding theory for the case of a slow uniform flow with $U_x = 1 \times 10^{-6}$ m/s and a slow rotational flow with $\Omega\Omega_x = 1 \times 10^{-6}$ rad/s. It is clearly seen that the numerical results of the force and the torque acting on the sphere are in good agreement with the theory.

4. Results and discussion

4.1. For the case of low Reynolds numbers with neglect of the inertial term

4.1.1. For the case of a uniform flow

We first consider the problem of a uniform flow past a cube in the x -direction, where the force and the torque acting on the cube are addressed for discussion. It is noted

Table 1. The force and torque acting on the spherical particle.

	The force acting on the spherical particle	The torque acting on the spherical particle
The theoretical values	$F_x^{\text{Theory}} = 6\pi\eta a U_x = 8.387 \times 10^{-11}$ [N]	$T_x^{\text{Theory}} = 8\pi\eta a^3 \Omega_x \approx 2.796 \times 10^{-15}$ [N·m]
The calculated values by ANSYS	$F_x^{\text{ANSYS}} = 8.392 \times 10^{-11}$ [N]	$T_x^{\text{ANSYS}} = 2.770 \times 10^{-15}$ [N·m]

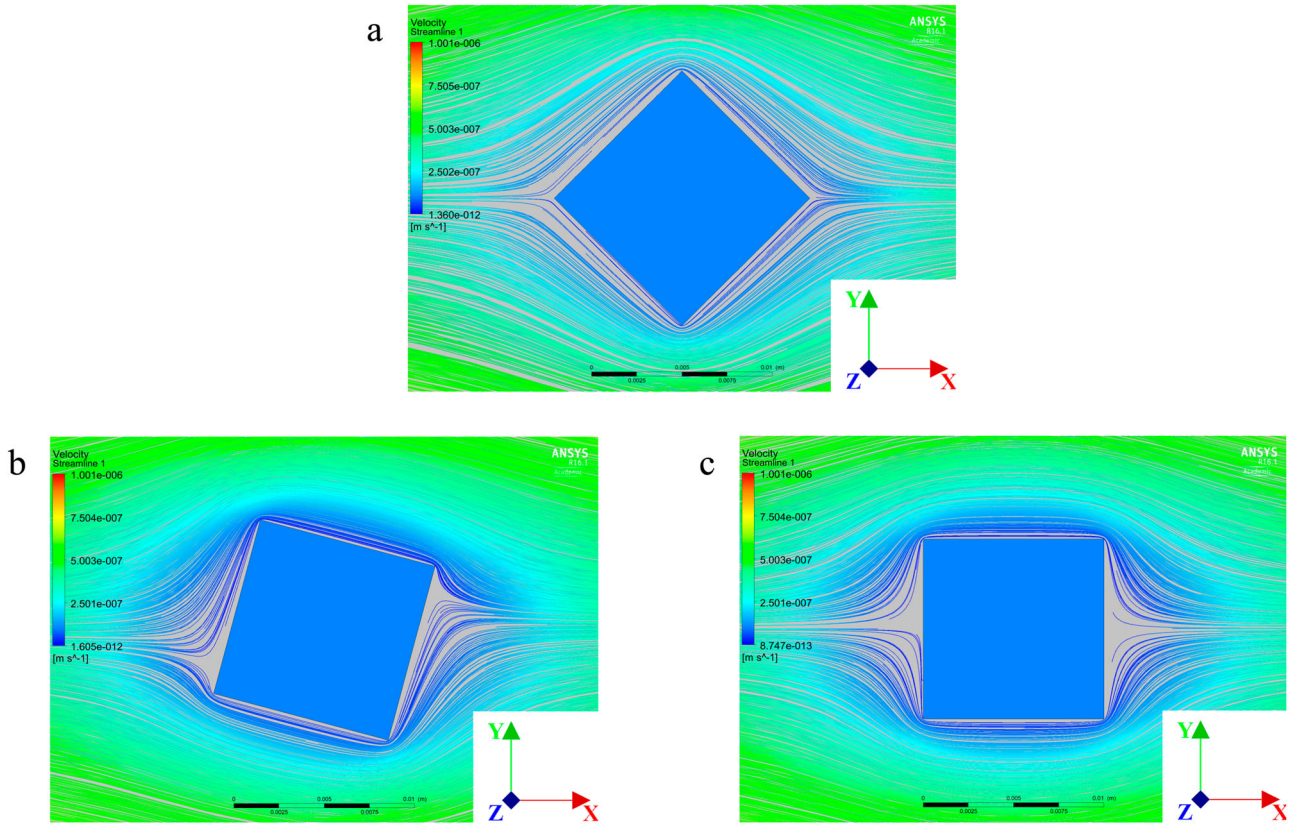


Figure 1. The flow field around a cube at each angle in the situation of a uniform flow field with a sufficiently low Reynolds number of $Re = 1.12 \times 10^{-2}$: (a) $\phi = 0^\circ$; (b) $\phi = 30^\circ$; (c) $\phi = 45^\circ$. A uniform flow past the cube is applied in the x-axis direction. In the cases of $\phi = 0^\circ$ and $\phi = 45^\circ$, the stream lines are approximately symmetric in the upstream and downstream with respect to the y-axis line.

that θ is the angle between the z-axis direction and the representative direction of the cube that is normal to a criterion face and ϕ is the rotation angle of a representative vertex about the z-axis measured from the x-axis toward the y-axis, therefore giving rise to $(\theta, \phi) = (0^\circ, 0^\circ)$ for the case of Figure 1(a).

Figure 1 shows results of the flow field around a cube with the fixed value of $\theta = 0^\circ$ with flow lines for three different cases of configuration $\phi = 0^\circ, 30^\circ$ and 45° . Figure 2 shows results for the components of the force and the torque acting on the cube. The results were obtained for a slow uniform flow $U_x = 1 \times 10^{-6}$ m/s with the resulting Reynolds number $Re = 1.12 \times 10^{-2}$. Since a slow uniform flow is addressed, the fluid tends to flow along the faces of the cube and into the region behind the cube without flow separation for all cases of the slant angle ϕ . From the flow lines shown in Figure 1(a) and 1(c), it is recognised that the flow lines are approximately symmetric about the y-axis line, which is well known for the case of a slow flow past a circular cylinder. From the results shown in Figure 2, it is seen that the force component F_x in the x-direction is certainly independent of the slant angle ϕ and exhibits a constant value $F_x \simeq 1.161 \times 10^{-10}$ N. In contrast, the

force components F_y and F_z are almost zero and thus negligible in comparison to the force component F_x along the uniform flow direction. Moreover, considering the torque $F_x (d/2) \simeq 0.58 \times 10^{-12}$ Nm evaluated with $F_x \simeq 1.161 \times 10^{-10}$ N as a criterion value for comparison, it is seen that all the torque components T_x , T_y and T_z are regarded as approximately zero. From these characteristics, we understand that a uniform flow gives rise to a force acting on the cube only in the uniform flow direction and does not induce a force in any other direction nor gives rise to an effective torque for any orientational configuration of the cube relative to the uniform flow.

We next consider the force and torque components for the case of a more general orientational configuration of the cube in a uniform flow. Figure 3 shows results of the flow field with flow lines for the case with the Reynolds number $Re = 1.12 \times 10^{-2}$, where the particle configuration is obtained firstly by rotating the cube about the Y-axis by $\theta = 54.74^\circ$ and then by rotating it about the new Z-axis by $\phi = 15^\circ$, giving rise to the general configuration of $(\theta, \phi) = (54.74^\circ, 15^\circ)$. It is noted that we have performed simulations for a variety of particle configurations, and the single case of the above configuration

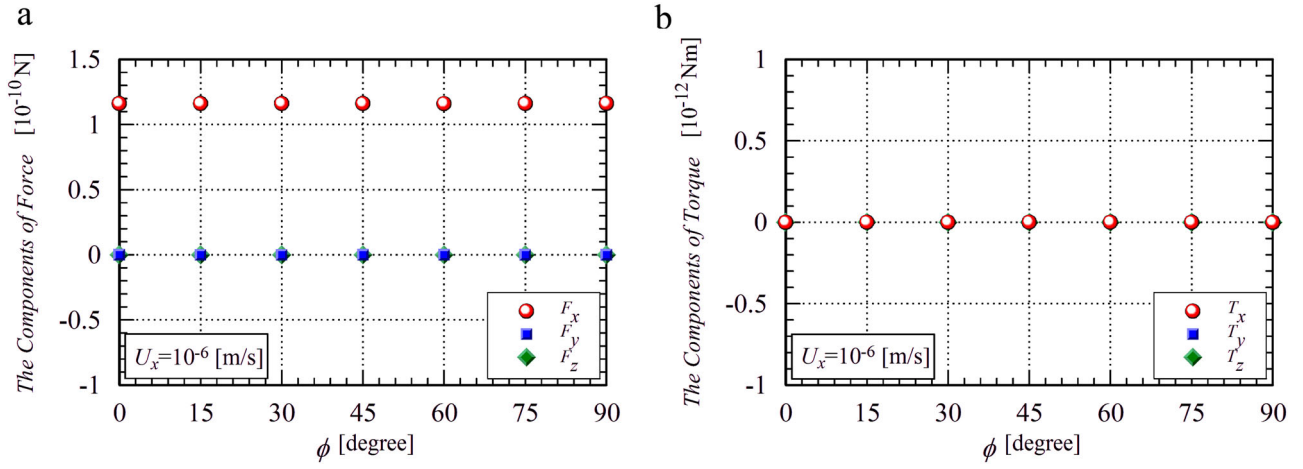


Figure 2. The force and torque acting on the cube for $Re = 1.12 \times 10^{-2}$: (a) the components of force; (b) the components of torque. The component of force F_x is independent of the configuration and therefore the value of F_x is almost constant. The components of force F_y and F_z are regarded as negligible in comparison to F_x .

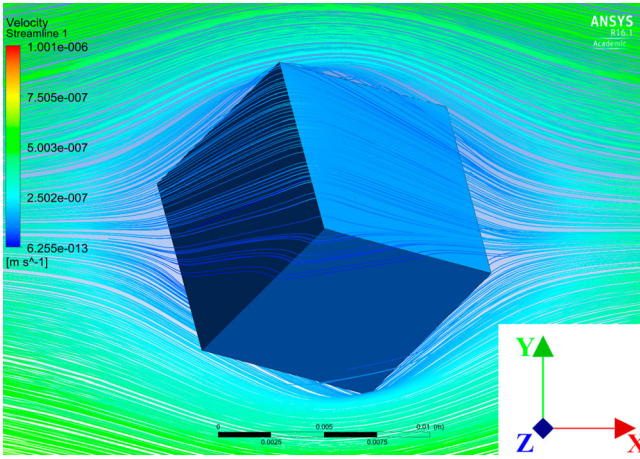


Figure 3. The flow field past the cube in the general orientational configuration in a 3D system for $Re = 1.12 \times 10^{-2}$. The fluid flows along the faces of the cube and flows away without flow separation.

has been employed to exemplify the following common conclusion. Table 2 shows the force components and the torque components acting on the cube, where the mean value of F_x for different ϕ values in the previous case shown in Figure 2(a), $\langle F_x \rangle$, is also shown for reference. As in the previous case, the fluid flows along the

faces of the cube and leaves the cube, where a flow separation or a vortex area does not arise behind the cube. Even in this general configuration, the force components F_y and F_z are sufficiently negligible and the force component F_x is equivalent to the value for the case shown in Figure 2. Moreover, as in the previous case, the torque components have meaningless values and are negligible when compared to the previous criterion value of F_x ($d/2$) $\simeq 0.58 \times 10^{-12}$ Nm.

From these results, we conclude that for the case of a slow uniform flow with a sufficiently small Reynolds number, the uniform flow gives rise to a force acting on the cube only in the uniform flow direction and does not induce a torque for any orientational configuration relative to the uniform flow direction.

We now describe the friction coefficients or the diffusion coefficients for a cube that are valid for a low Reynolds number with Re being much smaller than unity, using the above-mentioned flow characteristics. Even in the case of a cube, the resistance component matrix of $\mathbf{R}_a$ is independent of the orientational configuration i.e. the angles θ and ϕ , and is expressed as a diagonal matrix:

$$\mathbf{R}_a = \begin{pmatrix} \xi_{cube}^T & 0 & 0 \\ 0 & \xi_{cube}^T & 0 \\ 0 & 0 & \xi_{cube}^T \end{pmatrix}, \quad (10)$$

in which ξ_{cube}^T is the translational friction coefficient of the cube and is evaluated from Eq. (7) with the values of F_x and U_x as

$$\begin{aligned} \xi_{cube}^T &= \frac{\langle F_x \rangle}{U_x} = \frac{1.161 \times 10^{-10}}{1.0 \times 10^{-6}} = 1.161 \times 10^{-4} \\ &= 1.384 \times (3\pi\eta d). \end{aligned} \quad (11)$$

Table 2. The components of the force and the torque acting on the cube in the general orientational configuration in a 3D system for $Re = 1.12 \times 10^{-2}$.

F_x [N]	1.161×10^{-10}
F_y [N]	4.36×10^{-15}
F_z [N]	1.03×10^{-14}
T_x [Nm]	1.31×10^{-17}
T_y [Nm]	2.55×10^{-17}
T_z [Nm]	-2.94×10^{-18}
$\langle F_x \rangle$ [N]	1.161×10^{-10}

Since the torque is not induced by a slow uniform flow, the resistance coefficient matrix $\mathbf{R}_b$ is expressed as

$$\mathbf{R}_b = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (12)$$

This expression implies that, similar to a sphere, in the case of a cube a rotational motion is not induced by a uniform flow.

4.1.2. For the case of a rotational flow

We consider the problem of a rotational flow for the angular velocity $\Omega\Omega_x = 1 \times 10^{-6}$ rad/s, yielding a low Reynolds number of $Re = 1.12 \times 10^{-4}$, about the z -axis direction in order to clarify the characteristics of the resistance component matrices $\mathbf{R}_c$ and $\mathbf{R}_d$. Figure 4 shows results of the force components and the torque components acting on the cube for a variety of cases of the orientational angle ϕ with a fixed angle of $\theta = 0^\circ$. It is seen from Figure 4(a) that the comparison with a criterion value of $T_x/(d/2) \simeq 1.43 \times 10^{-12}$ N gives rise to the conclusion that the force components F_x , F_y and F_z may be regarded as negligibly small. Hence, it implies that a rotational flow field does not induce a force acting on the cube in a specific direction. Moreover, only the torque T_x exhibits a meaningful value, independent of the orientational angle ϕ and the other components T_y and T_z are negligible in comparison to the torque component T_x .

We next consider the force and torque components for the case of a more general orientational configuration of the cube in a rotational flow with $(\theta, \phi) = (54.74^\circ, 15^\circ)$ for the same Reynolds number of $Re = 1.12 \times 10^{-4}$. Table 3 shows results of the force components and the torque components acting on the cube, where the value

Table 3. The components of the force and the torque acting on the cube in a more general particle configuration in a 3D system for $Re = 1.12 \times 10^{-4}$.

F_x [N]	-2.23×10^{-16}
F_y [N]	1.19×10^{-14}
F_z [N]	3.94×10^{-15}
T_x [Nm]	7.140×10^{-15}
T_y [Nm]	1.95×10^{-18}
T_z [Nm]	1.43×10^{-18}
$\langle T_x \rangle$ [Nm]	7.134×10^{-15}

of $\langle T_x \rangle$, equivalent to T_x , from the previous case shown in Figure 4 is also shown for reference. It is seen that the torque T_x may be regarded as being equivalent to that in the previous case of a specific orientational configuration shown in Figure 4(b). The torque components T_y and T_z are seen to be sufficiently small and thus are negligible in comparison to the value of T_x . This implies that the torque acting on the cube in a rotational flow is determined only by the torque about the direction of the rotational flow. By the comparison with a criterion value of $T_x/(d/2) \simeq 1.43 \times 10^{-12}$ N with $T_x = 7.14 \times 10^{-15}$ Nm, all the force components may be considered negligible in comparison to this criterion force.

From these results, we understand that in the case of a slow rotational flow with a sufficiently small Reynolds number of $Re = 1.12 \times 10^{-4}$, the flow induces only a torque about the direction of the rotational flow field and does not induce a force acting on the cube for the case of any orientational configuration. That is, the rotational motion does not induce a translational motion of the cube for this rotational flow field.

We now relate the above-mentioned characteristics in a rotational flow to the friction coefficients or the diffusion coefficients of a cube that are valid for a low Reynolds

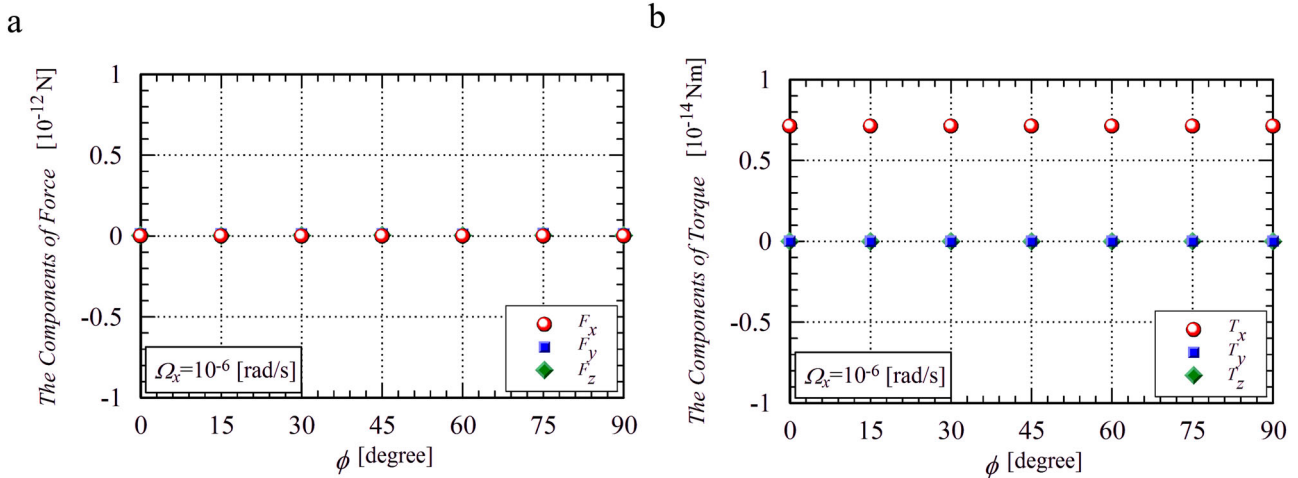


Figure 4. The force and torque acting on the cube for $Re = 1.12 \times 10^{-4}$: (a) the components of force; (b) the components of torque. The component of torque T_x is independent of the configuration and almost constant. The components of torque T_y and T_z are regarded as negligible in comparison to T_x .

number with Re being much smaller than unity. Since the rotational motion does not induce a translational motion of the cube and only the torque component about the direction of the rotational flow field has a meaningful value, the resistance component matrix of $\mathbf{R}_d$ is independent of the orientational configuration and as in the case of a uniform flow field, it is expressed as a diagonal matrix as follows:

$$\mathbf{R}_d = \begin{pmatrix} \xi_{cube}^R & 0 & 0 \\ 0 & \xi_{cube}^R & 0 \\ 0 & 0 & \xi_{cube}^R \end{pmatrix}, \quad (13)$$

in which ξ_{cube}^R is the rotational friction coefficient of a cube and is evaluated using the values of the torque component T_x shown in Table 3 and the angular velocity Ω_x of the rotational flow field as

$$\begin{aligned} \xi_{cube}^R &= \frac{\langle T_x \rangle}{\Omega_x} = \frac{7.134 \times 10^{-15}}{1.0 \times 10^{-6}} = 7.134 \times 10^{-9} \\ &= 2.552 \times (\pi \eta d^3). \end{aligned} \quad (14)$$

Since the rotational motion does not induce a force, the resistance component matrix $\mathbf{R}_c$ is written as

$$\mathbf{R}_c = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (15)$$

From the characteristics of the friction coefficients of a cube that have been obtained from both a uniform flow and a rotational flow, we conclude that there is no coupling between the translation motion and the rotational motion in the situation of the Reynolds number being sufficiently smaller than unity, and therefore the motion of the cube need only to be characterised by two friction coefficients, i.e. the translational friction and rotational friction coefficients shown in Eqs. (11) and (14), respectively, as in the case of a sphere.

4.2. For the case of non-small Reynolds numbers with consideration of the inertial term

In this section we consider a uniform flow problem where the Reynolds number is larger than unity and thus the coupling between the translational motion and the rotational motion of a cube is expected, resulting in the possibility of the resistance component matrices $\mathbf{R}_b$ and $\mathbf{R}_c$ being non-negligible. To this end, the uniform velocity is taken as $U_x = 1 \times 10^{-2}$ m/s with $Re = 1.12 \times 10^2$, and as in the previous case shown in Figure 1, a cube is set in the orientational configuration of $\theta = 0^\circ$ for several cases of the angle ϕ .

Figure 5 shows results of the flow field with flow lines for the two cases of the orientational configurations $\phi = 0^\circ$ and $\phi = 45^\circ$. Figure 6 shows the dependence of the force components and the torque components on the configuration angle ϕ . The flow lines shown in Figs. 5(a) and (b) clearly exhibit vortex rings implying a pair of vortices in the 2D plane behind the cube that not symmetric with respect to the y -axis. In significant contrast to the case of a slow uniform flow field, the force component F_x shown in Figure 6(a) is not constant but strongly dependent on the slant angle ϕ . Moreover, the value of the force component F_y is no longer negligible in comparison to F_x and exhibits a meaningful value. Also with respect to the torque components, the results in Figure 6(b) clearly show that the torque component T_x is induced at a meaningful level in the non-symmetric orientational configurations such as $\phi = 15^\circ, 30^\circ, 60^\circ$ and 75° . Although not shown as a result, a coupling behaviour between the translational motion and the rotational motion has also been recognised for the case of a different value of the angle θ . A noticeable feature in Figure 6(a) is that the configuration of $\phi = 0^\circ$ and $\phi = 45^\circ$ gives rise to a maximum and a minimum value of the force component F_x ,

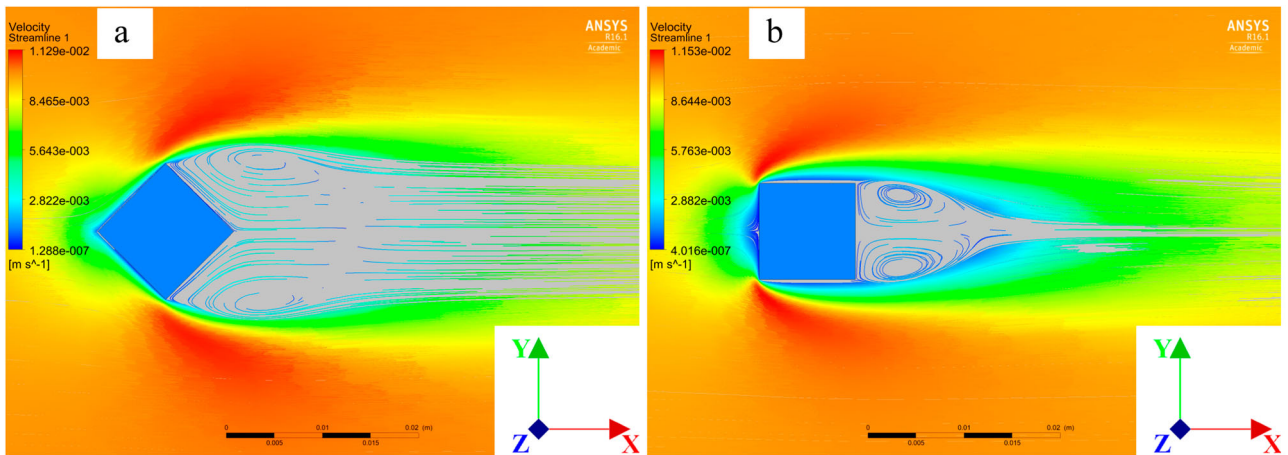


Figure 5. The flow field around a cube oriented in a uniform flow field with a large Reynolds number of $Re = 1.12 \times 10^2$: (a) $\phi = 0^\circ$; (b) $\phi = 45^\circ$. A uniform flow past the cube is applied in the x -axis direction. The pair of the vortices appear in the flow behind the cube.

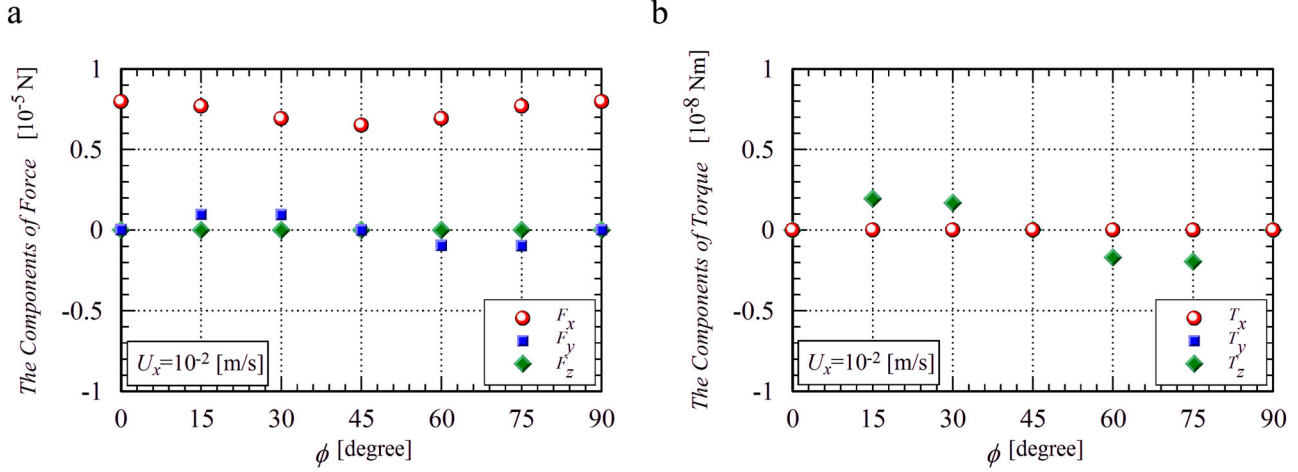


Figure 6. The force and torque acting on the cube for $Re = 1.12 \times 10^2$: (a) The component of force; (b) The component of torque. The components of force F_x and F_y depend on the configuration and the component of force F_z and are regarded as negligible in comparison to F_x . The component of torque T_z is also acting on the cube.

respectively, which may be related to the projected area of the cube on the plane normal to the uniform flow direction in the case of a non-slow uniform flow.

From the above flow properties of the cube, we now consider the characteristics of the resistance component matrices. Since the coupling between the translational motion and the rotational motion is no longer negligible in a non-slow uniform flow past a cube, the resistance component matrices are not able to be expressed in a simple form, such as a diagonal matrix, and they are also dependent on the orientational configuration i.e. on the orientational angles θ and ϕ . Hence, each translational friction coefficient is dependent on the orientational angles and the other axis components, whilst the rotational friction coefficients are dependent on the other friction coefficients in a similar manner. Thus, the resistance component matrix $\mathbf{R}_a = \mathbf{R}_a(\theta, \phi)$ is expressed as

$$\mathbf{R}_a(\theta, \phi) = \begin{pmatrix} \xi_{cube,11}^T(\theta, \phi) & \xi_{cube,12}^T(\theta, \phi) & \xi_{cube,13}^T(\theta, \phi) \\ \xi_{cube,21}^T(\theta, \phi) & \xi_{cube,22}^T(\theta, \phi) & \xi_{cube,23}^T(\theta, \phi) \\ \xi_{cube,31}^T(\theta, \phi) & \xi_{cube,32}^T(\theta, \phi) & \xi_{cube,33}^T(\theta, \phi) \end{pmatrix}, \quad (16)$$

in which $\xi_{cube,11}^T(\theta, \phi), \xi_{cube,12}^T(\theta, \phi), \dots, \xi_{cube,33}^T(\theta, \phi)$ are a friction coefficient where for example, $\xi_{cube,12}^T(\theta, \phi)$ corresponds to the case where the motion of the cube in the y -axis direction induces the force in the x -axis direction. It is noted that these friction coefficients are not constant but vary dependent on the orientational configuration of the cube, although in the case of a slow uniform flow this resistance component matrix is diagonal and its

components are constant and independent of the orientational configuration. Similarly, the resistance component matrix $\mathbf{R}_b = \mathbf{R}_b(\theta, \phi)$ is expressed as

$$\mathbf{R}_b(\theta, \phi) = \begin{pmatrix} \xi_{cube,11}^{TR}(\theta, \phi) & \xi_{cube,12}^{TR}(\theta, \phi) & \xi_{cube,13}^{TR}(\theta, \phi) \\ \xi_{cube,21}^{TR}(\theta, \phi) & \xi_{cube,22}^{TR}(\theta, \phi) & \xi_{cube,23}^{TR}(\theta, \phi) \\ \xi_{cube,31}^{TR}(\theta, \phi) & \xi_{cube,32}^{TR}(\theta, \phi) & \xi_{cube,33}^{TR}(\theta, \phi) \end{pmatrix}. \quad (17)$$

in which $\xi_{cube,11}^{TR}(\theta, \phi), \xi_{cube,12}^{TR}(\theta, \phi), \dots, \xi_{cube,33}^{TR}(\theta, \phi)$ are the synergistic friction coefficients relating the forces acting on the cube to the angular velocities of the rotational motion. In the above case of a small Reynolds number all these friction coefficients would have been zero.

We next consider a rotational flow for the angular velocity of $\Omega \Omega_x = 1$ rad/s with the Reynolds number $Re = 1.12 \times 10^2$ in order to evaluate the resistance component matrices $\mathbf{R}_c$ and $\mathbf{R}_d$. Figure 7 shows the dependence of the forces and the torques acting on the cube on the orientational angle ϕ with a fixed angle $\theta = 0^\circ$. From the result shown in Figure 7(a), it is seen that all the force components exhibit significantly small values and may be regarded as negligible, as in the case of a small Reynolds number shown in Figure 4(a). This clearly implies that even in the case of non-small Reynolds numbers the rotational motion of the cube does not induce a translational motion. Moreover, the torque component T_x about the direction of the angular velocity of the rotational flow field exhibits sufficiently large values and a slight dependence on the orientational angle ϕ , whereas the other torque components T_y and T_z have negligibly small values in comparison to the values of T_x . Although the

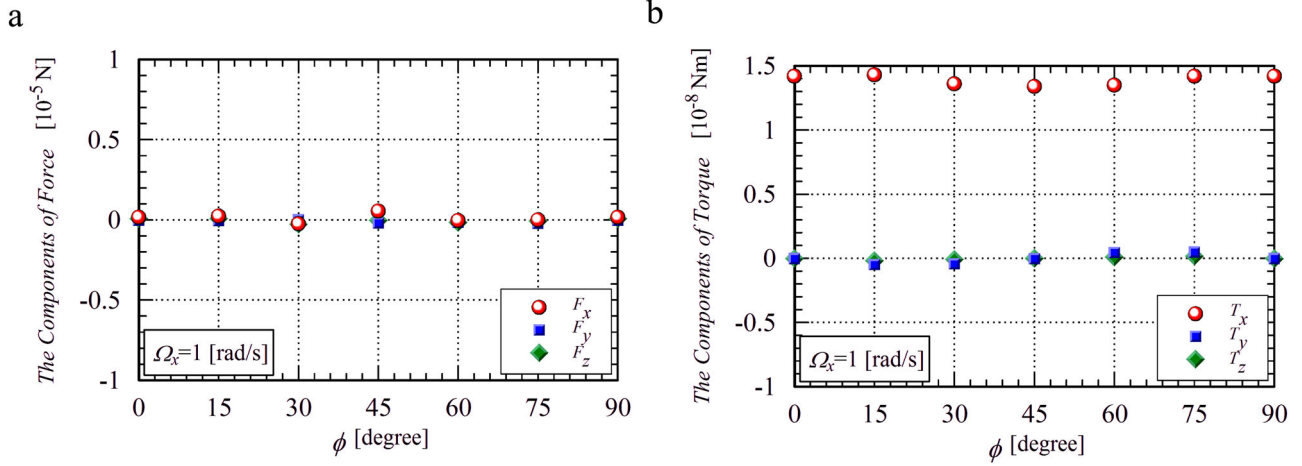


Figure 7. The force and torque acting on the cube for $Re = 1.12 \times 10^2$: (a) the force components; (b) the torque components. It is seen that the force acting on the cube is not induced by the rotational flow field. Only the torque component T_x about the angular velocity direction is induced but the other torque components T_y and T_z may be regarded as negligible in comparison to the torque component T_x .

dependence on the angle θ is not shown here as a figure, these characteristics regarding the torque components are also valid. We therefore understand that the rotational flow field induces only a torque about the direction of the angular velocity of the rotational flow field and the other torque components may be regarded as negligible.

From the above-mentioned characteristics of the forces and torques acting on the cube in a rotational flow field, we see that the rotational motion does not induce a translational motion and that the rotational motion about the angular velocity vector of the rotational flow field is significantly induced in the case of non-small Reynolds numbers. Similar to the resistance component matrices $\mathbf{R}_a$ and $\mathbf{R}_b$ shown in Eqs. (16) and (17), the matrices $\mathbf{R}_c$ and $\mathbf{R}_d$ are expressed in general form as

$$\begin{aligned} \mathbf{R}_d(\theta, \phi) &= \begin{pmatrix} \xi_{cube,11}^R(\theta, \phi) & \xi_{cube,12}^R(\theta, \phi) & \xi_{cube,13}^R(\theta, \phi) \\ \xi_{cube,21}^R(\theta, \phi) & \xi_{cube,22}^R(\theta, \phi) & \xi_{cube,23}^R(\theta, \phi) \\ \xi_{cube,31}^R(\theta, \phi) & \xi_{cube,32}^R(\theta, \phi) & \xi_{cube,33}^R(\theta, \phi) \end{pmatrix}, \end{aligned} \quad (18)$$

$$\begin{aligned} \mathbf{R}_c(\theta, \phi) &= \begin{pmatrix} \xi_{cube,11}^{RT}(\theta, \phi) & \xi_{cube,12}^{RT}(\theta, \phi) & \xi_{cube,13}^{RT}(\theta, \phi) \\ \xi_{cube,21}^{RT}(\theta, \phi) & \xi_{cube,22}^{RT}(\theta, \phi) & \xi_{cube,23}^{RT}(\theta, \phi) \\ \xi_{cube,31}^{RT}(\theta, \phi) & \xi_{cube,32}^{RT}(\theta, \phi) & \xi_{cube,33}^{RT}(\theta, \phi) \end{pmatrix}, \end{aligned} \quad (19)$$

in which $\xi_{cube,11}^R(\theta, \phi), \dots, \xi_{cube,33}^R(\theta, \phi), \xi_{cube,11}^{RT}(\theta, \phi), \dots, \xi_{cube,33}^{RT}(\theta, \phi)$ are the rotational friction coefficients

relating the torques acting on the cube to the translational and angular velocities. By taking into account the results discussed above, the resistance matrix $\mathbf{R}_c$ may be regarded as $\mathbf{R}_c = \mathbf{0}$ with approximately all zero components. The resistance component matrix $\mathbf{R}_d$ may be expressed by an equation similar to Eq. (13). However, it should be noted that this assumption is valid for the present Reynolds number, and thus for a larger Reynolds numbers these resistance component matrices are expected to be written in a more general form.

5. Application to the Brownian dynamics simulation method

The final objective of the present study has been to develop a Brownian dynamics simulation method for a suspension composed of cubic particles in order to elucidate the dynamic characteristics of the suspension. Hence, we now consider the basic equations of the translational and rotational motion of cubic particles by taking into account the characteristics of the friction or the diffusion coefficients that were obtained in the previous sections. We treat a cubic magnetic particle with a micron or smaller order dimension and therefore the Stokes approximation is applicable, whereby physical problems for the case of a slow Reynolds number are treated. A discussion above has given rise to the conclusion that there is no coupling between the translation motion and the rotational motion of a cubic particle system as is found in the case of a spherical particle system. This implies that we are able to treat the translational motion and the rotational motion separately in a Brownian dynamics simulation. From the results obtained by ANSYS for the case of $d = 1$ cm, we

have shown that the translational and rotational friction coefficients are written in mathematical expressions as $\xi_{cube}^T = 1.384 \times (3\pi\eta d)$ and $\xi_{cube}^R = 2.552 \times (\pi\eta d^3)$. In the situation of $Re \ll 1$, these expressions hold independent of the values of d and η , and are thus applicable to cubic particles with the size of $d \lesssim 1 \mu\text{m}$ that perform Brownian motion. Actually, although not shown as figures, our other ANSYS numerical analyses for particles with the sizes of $d = 1 \text{ mm}$ and $1 \mu\text{m}$ have given rise to the same relationships shown in Eqs. (11) and (14).

If we employ the notation $\mathbf{r}$ for the particle position and $\mathbf{n}$ for the direction of the magnetic moment, the basic equations of the particle position and the magnetic moment direction employed in a Brownian dynamics simulation are written as [23,24]

$$\begin{aligned} \mathbf{r}(t + \Delta t) &= \mathbf{r}(t) + \frac{1}{kT} D_{cube}^T \mathbf{F}^P(t) \Delta t + \Delta \mathbf{r}^B, \\ \mathbf{n}(t + \Delta t) &= \mathbf{n}(t) + \frac{1}{kT} D_{cube}^R \mathbf{T}^P(t) \times \mathbf{n} \Delta t + \Delta \boldsymbol{\phi}^B, \end{aligned} \quad (20)$$

in which k is Boltzmann's constant, T is the liquid temperature and Δt is the time interval in a simulation. The random displacements inducing the translational and rotational Brownian motion, $\Delta \mathbf{r}^B$ and $\Delta \boldsymbol{\phi}^B$ respectively, are described in detail in the reference [23,24] and therefore the expressions are not shown here. D_{cube}^T and D_{cube}^R are the diffusion coefficients and are related to the friction coefficients ξ_{cube}^T and ξ_{cube}^R by the expressions shown in Eq. (6). Although these friction coefficients ξ_{cube}^T and ξ_{cube}^R have already been shown in Eqs. (11) and (14), they are rewritten here in a different form in order to relate them to the friction coefficients of the sphere with the mean diameter of the inscribed and the circumscribed sphere. The translational and rotational friction coefficients $\xi_{sphere}^{T(mean)}$ and $\xi_{sphere}^{R(mean)}$ of the sphere with the mean diameter of the inscribed and the circumscribed sphere are written as

$$\begin{aligned} \xi_{sphere}^{T(mean)} &= \frac{1 + \sqrt{3}}{2} \xi_{sphere}^T, \\ \xi_{sphere}^{R(mean)} &= \left(\frac{1 + \sqrt{3}}{2} \right)^3 \xi_{sphere}^R, \end{aligned} \quad (22)$$

in which ξ_{sphere}^T and ξ_{sphere}^R are the friction coefficients for the inscribed sphere with diameter d . Using these mean friction coefficients, the translational and rotational friction coefficients ξ_{cube}^T and ξ_{cube}^R for the cube are written as

$$\xi_{cube}^T = \alpha_{cube}^T \xi_{sphere}^{T(mean)}, \quad \xi_{cube}^R = \alpha_{cube}^R \xi_{sphere}^{R(mean)}, \quad (23)$$

in which α_{cube}^T and α_{cube}^R are the modification coefficients and they are given as $\alpha_{cube}^T = 1.013$ and $\alpha_{cube}^R =$

1.001 from Eqs. (11) and (14). The values of α_{cube}^T and α_{cube}^R are nearly equivalent to unity and therefore the friction coefficients of the cube are roughly equivalent to those of the translational and rotational friction coefficients of the sphere with the mean diameter of the inscribed and the circumscribed sphere. It is noted that Eqs. (20), (21) and (23) are non-dimensionalized using the corresponding representative values in order to perform the Brownian dynamics simulations. Since the friction coefficients ξ_{cube}^T and ξ_{cube}^R of the cube are divided by those of the sphere with diameter d , we treat the following non-dimensionalized friction coefficients $\xi_{cube}^{T*} (= \xi_{cube}^T / (3\pi\eta d))$ and $\xi_{cube}^{R*} (= \xi_{cube}^R / (\pi\eta d^3))$ in the present simulations:

$$\xi_{cube}^{T*} = \alpha_{cube}^T \frac{1 + \sqrt{3}}{2}, \quad \xi_{cube}^{R*} = \alpha_{cube}^R \left(\frac{1 + \sqrt{3}}{2} \right)^3. \quad (24)$$

It is noted that the non-dimensionalized friction coefficients ξ_{cube}^{T*} and ξ_{cube}^{R*} are constants as $\xi_{cube}^{T*} = 1.384$ and $\xi_{cube}^{R*} = 2.552$ and independent of the size of the cubic particle.

We may finally conclude that in the case of a low Reynolds number being much smaller than unity the Brownian dynamics simulation method for a spherical particle system is straightforwardly applicable to a suspension composed of cubic particles by employing the above-mentioned friction coefficients of the cube shown in Eq. (23).

In order to verify the validity of the Brownian dynamics simulations employing the above proposed diffusion coefficients, we briefly discuss the aggregate structure of magnetic cubic particles by focusing on the radial distribution function, although a detailed discussion is proposed for a future paper. It is noted that the radial distribution function is a reasonable index for a quantitative discussion on the internal structure of particle aggregates since it is significantly sensitive to the form of an aggregate structure. For comparison, the corresponding Monte Carlo result is employed as a criterion result since Monte Carlo is a fully developed simulation technique and results obtained by this simulation method can be regarded as a corresponding to a theoretical solution. Since the Brownian dynamics and Monte Carlo simulation methods for magnetic particles are sufficiently explained in several references [16,23,24], we here only describe the non-dimensional parameters that characterise the aggregation phenomenon of the magnetic particles. Two non-dimensional parameters, ξ and λ , characterise the present aggregation phenomenon in a suspension of cubic magnetic particles in an applied magnetic field, where ξ and λ imply the strengths of the

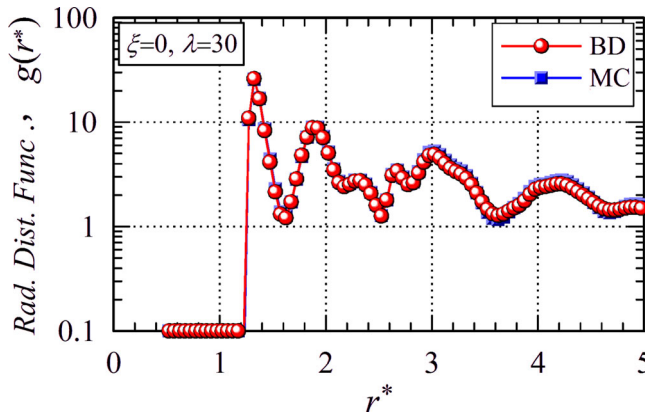


Figure 8. Radial distribution functions for the magnetic interaction strength $\lambda = 30$ in no applied magnetic field. The result obtained by Brownian dynamics simulations is in significantly good agreement with the corresponding Monte Carlo result [14].

magnetic particle-field and the magnetic particle-particle interaction, respectively. If the value of one these quantities is much larger than unity, then its effect is much more dominant than the influence of the thermal or random motion of magnetic particles. Figure 8 shows the radial distribution functions obtained from simulations of a suspension with a magnetic interaction strength $\lambda = 30$ in the absence of an applied magnetic field $\xi = 0$, with a volumetric fraction $\phi_V = 0.05$ and a particle number $N = 490$. From Figure 8, it is evident that the result obtained by a Brownian dynamics simulation is in very good agreement with the corresponding Monte Carlo result in regard to the shape and the height of each peak. This implies the aggregate structure of the magnetic cubic particles is in very good agreement with that obtained by a Monte Carlo simulation. This evidently verifies that the diffusion coefficients shown in Eq. (11) and (14) are able to be used for the case of a cubic particle in Brownian dynamics simulations shown in Eqs. (20) and (21).

6. Conclusion

In the present study, we have considered a flow problem around a cube by a numerical analysis approach in order to clarify the characteristics of the translational and rotational friction or diffusion coefficients of a cubic particle that are a necessary requirement for implementing particle-based simulations of a cubic particle suspension. In concrete, by estimating the force and torque acting on the cube in the two cases of a uniform flow field and a rotational flow field, we have discussed whether or not the translational and the rotational motion have a coupling relationship in the two regimes of a slow flow and a non-slow flow with the Reynolds number being respectively much smaller and larger than unity. From

the characteristics of the friction coefficients of a cube that have been obtained from both a uniform flow and a rotational flow, we may understand that there is no coupling between the translation motion and the rotational motion in the situation of the Reynolds number being sufficiently smaller than unity. Hence, the motion of the cube can be characterised with only two friction coefficients, i.e. the translational friction and the rotational friction coefficients, as in the case of a spherical particle. In contrast, in the case of a non-slow flow field with the Reynolds number larger than unity, a coupling characteristic between the translational motion and the rotational motion is certainly recognised and therefore the interaction with the ambient fluid is characterised by a variety of friction coefficients but a variety of friction coefficients that correlate the translational and the rotational motion. We have shown the expressions of the friction or diffusion coefficients relative to the corresponding values of an inscribed and a circumscribed sphere. Employment of these translational and rotational diffusion coefficients for a cube enables the implementation of Brownian dynamics simulations for a suspension composed of cubic particles as in the case of a spherical particle suspension in order to analyse the dynamics characteristics of a cubic particle suspension.

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Disclosure statement

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